

# Predator–prey interactions as drivers of cognitive evolution

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## Abstract

How and why cognitive phenotypes vary between and within species remain contested. The cognitive demands associated with social interactions and environmental variability are hypothesized to be the predominant factors governing cognitive evolution. However, these factors fail to account for a large fraction of variation in animal cognition. Evidence is accumulating that interactions between predators and prey can drive cognitive evolution, although exploring this link has been constrained by methodological barriers and is often overlooked. In this Perspective, we outline how predator–prey interactions produce cognitive variation and drive cognitive evolution. We formalize this as the predatory intelligence hypothesis (PIH), which posits that the cognitive challenges associated with predator–prey interactions drive a cognitive co-evolutionary arms race between predators and prey, and that the result of this race is bidirectional enhancements across a suite of cognitive traits. We provide a series of predictions based on the PIH, and detail avenues for future research that will help to clarify the role of predation in cognitive evolution.

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## Introduction

Cognition – the processes by which individual animals acquire, process, store and act on information from the environment<sup>1</sup> – varies greatly among individuals, populations and species, but why it varies remains a topic of debate. One of the predominant hypotheses for cognitive evolution, the social intelligence hypothesis (SIH), posits that the challenges associated with social life select for enhanced cognitive abilities. The pressures of social living favour enhanced cognitive abilities as they require maintaining and coordinating relationships, monitoring group members and outsiders, identifying cooperative partners and outwitting conspecifics in Machiavellian interactions<sup>2,3</sup>. Although the SIH has empirical and theoretical support, more than 40% of cognitive variation remains unaccounted for<sup>4</sup>. By contrast, the ecological intelligence hypothesis (EIH)<sup>5,6</sup> posits that cognitive evolution is driven by the challenges of food acquisition, environmental heterogeneity and uncertain climatic conditions. It is hypothesized that these challenges select for cognitive traits such as spatial memory and innovation, which promote the ability to recognize and respond to environmental cues<sup>7–11</sup>. Similar to the SIH, research testing the EIH has produced equivocal findings, with conflicting results across space, time and taxa<sup>12–14</sup>.

Another long-posed driver of cognitive evolution is the interactions between predators and prey<sup>15–18</sup>. The role of predator–prey interactions in cognitive evolution has historically been described as an “ecological theory yet to be fully evaluated”<sup>15</sup>, which still largely holds true today. Other researchers have considered predation to be a confounding factor when testing how sociality shapes brain size<sup>19</sup>, making it imperative to consider when testing social drivers of cognitive evolution. Some elements of the EIH incorporate the interplay between hunting behaviour and cognitive evolution<sup>20,21</sup>, positing that the need to acquire resources (prey) drove the emergence of enhanced predator cognitive abilities. However, predation is largely peripheral to the central predictions of the EIH, which focus on vegetative dietary diversity and environmental variability<sup>6,14</sup> and overlook how predation selects for cognitive ability in prey. In predator–prey interactions, animals must rely not only on direct behavioural cues but also on indirect cues, some of which are deliberately concealed or designed to mislead. Reliance on cryptic and indirect cues, and rapid response to direct cues (for example, spotting a predator), present a set of unique and complex cognitive demands that are not adequately captured by the EIH.

Rapidly accumulating evidence indicates that predator–prey interactions drive cognitive variation via the cognitive challenges that accompany the processes of both prey acquisition and predator avoidance. For example, prey rely on recognizing and associating cues with danger, recalling risky places and times, and reacting quickly to actual predator encounters. Predator avoidance can involve coordinating group movements and adjusting behaviour via social learning (Fig. 1). In some species, such as pheasants (*Phasianus colchicus*) and African striped mice (*Rhabdomys pumilio*), enhanced cognitive abilities are a clear driver of prey survival through avoidance of predation<sup>22,23</sup>. Additionally, exposure to intense predation pressure drives increases in brain and telencephalon size in guppies (*Poecilia reticulata*)<sup>24</sup>. Female guppies with larger brains are better able to assess predation risk than those with smaller brains by reducing the time required to gather and assess information about a predator before deciding upon a response<sup>25</sup>. Furthermore, at a macroevolutionary scale, prey with morphological anti-predator defences (such as body armour or spines) tend to have smaller brains, suggesting that physical defences reduce the need for the advanced cognitive abilities required for behavioural predator avoidance<sup>26</sup>.

In this Perspective, we provide a conceptual synthesis that unites concepts from the study of cognitive evolution and predator–prey ecology to establish the predatory intelligence hypothesis (PIH), which predicts that predation is an important driver of variation in cognitive ability at the individual, developmental and evolutionary levels. The PIH posits that the cognitive challenges associated with interactions between predators and their prey can create a co-evolutionary arms race that promotes bidirectional enhancements across several cognitive traits. Finally, we present key predictions and propose approaches to testing them, that span from individuals to species and from deep time to modern day, to evaluate predation’s role in cognitive evolution and the maintenance of variation in cognitive abilities.

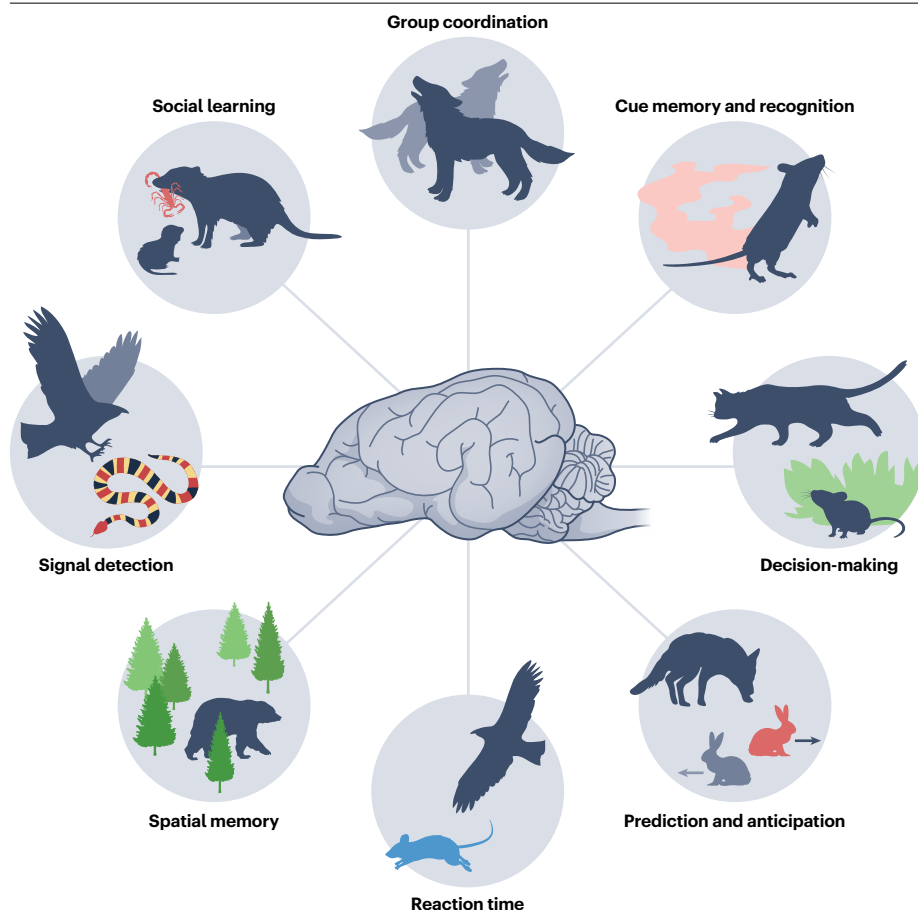
## The predatory intelligence hypothesis

Predator–prey interactions shape the lives of most animals, as a necessary interaction in the acquisition of food for predators or the avoidance of death for prey. Predation and predator avoidance each have their own cognitive demands and selective pressures. In this section, we establish the basis of the PIH, connecting predator–prey interactions and their associated cognitive demands.

## Linking predator–prey interactions and cognition

Predators are under strong selection to be efficient hunters, and prey are under strong selection to avoid predation, although there is substantial asymmetry in this selection pressure<sup>27</sup>. Introducing predators and prey into novel environments can provide evidence of this evolutionary pressure<sup>28,29</sup>. For instance, when predatory curly-tailed lizards (*Leiocephalus carinatus*) were introduced to small Caribbean islands, their potential prey, brown anoles (*Anolis sagrei*), became less bold and more arboreal, thereby substantially reducing their risk of predation<sup>30</sup>. For predators, novel prey can drive rapid behavioural<sup>31</sup> and morphological evolution<sup>28</sup>, promoting traits that facilitate hunting success. For example, when island apple snails (*Pomacea maculata*) were introduced to North America, snail kites (*Rostrhamus sociabilis*) rapidly evolved larger bodies and bills in response to this new prey source<sup>28</sup>. Predator–prey theory suggests that the outcomes of interactions depend on the capacity of each species to assess the other’s physical condition, behaviour and intent, adjust behaviour based on past experience and, ultimately, make decisions that maximize the probability of survival<sup>32,33</sup>. Perceiving and interpreting information about risk and reward are at the forefront of this evolutionary arms race. Thus, large components of predatory and anti-predator behaviours require skills and sensorimotor capabilities that are dependent on a substantial amount of information processing<sup>34</sup>. Enhanced executive function – the ability to rapidly integrate information and respond appropriately – should therefore offer a clear advantage in these life and death encounters between predators and prey<sup>5,35,36</sup>.

Cognitive ability can be an important predictor of prey survival and fitness<sup>4,13,37–39</sup>. African striped mice with enhanced cognitive performance have faster reaction times to predator cues and survive longer in the wild compared with those with poorer cognitive performance<sup>23,39</sup>. Additionally, the combined cognitive performance score (calculated from performance in problem-solving, spatial memory, inhibitory control and causal understanding tasks) of grey mouse lemurs (*Microcebus murinus*) is correlated with survival in the wild, where predators are the primary source of mortality<sup>38</sup>. Pheasants that exhibit enhanced spatial memory are better at avoiding predators<sup>22</sup>. Neuroanatomical proxies (such as total brain size) for cognitive abilities can also influence anti-predator behaviour. Female guppies with larger brains take



**Fig. 1 | Cognitive challenges of predator–prey interactions.** Predator–prey interactions remain largely unexplored as a driver of cognitive variation and evolution despite sharing key mechanisms with other predominant hypotheses explaining cognitive evolution. Predators and prey must learn to remember and recognize the olfactory, visual and auditory cues of their counterparts<sup>33,68</sup>; prey must make decisions about when to hide, flee and forage<sup>111</sup>, whereas predators must make decisions about which prey to attack and the manner and timing in which to do so<sup>112</sup>. Each member of the dyad must evaluate the cues of the other to make an informed behavioural decision; prey can be selected for their reaction times<sup>23</sup>, and the same is probably true for predators; also, predators and prey should remember the locations of the other member of the interaction to either avoid or force an interaction<sup>22</sup>. Although only documented in prey<sup>94,113,114</sup>, the ability to socially learn predatory and anti-predator behaviours is central to predator–prey interactions<sup>59</sup>; and group coordination for hunting prey and avoiding predation presents a key social challenge that is shaped by predation landscapes<sup>115–117</sup>.

less time to recognize and avoid predators in laboratory experiments<sup>25</sup>. Further, larger brains are considered to make predators better hunters owing to improved cognitive abilities<sup>40</sup>. For example, increases in the brain size of hominids in Africa corresponded with the decline of predatory megafauna, suggesting that improved cognitive performance might have allowed early humans to eradicate their competitors and predators<sup>40</sup>, although this hypothesis is contested<sup>41</sup>.

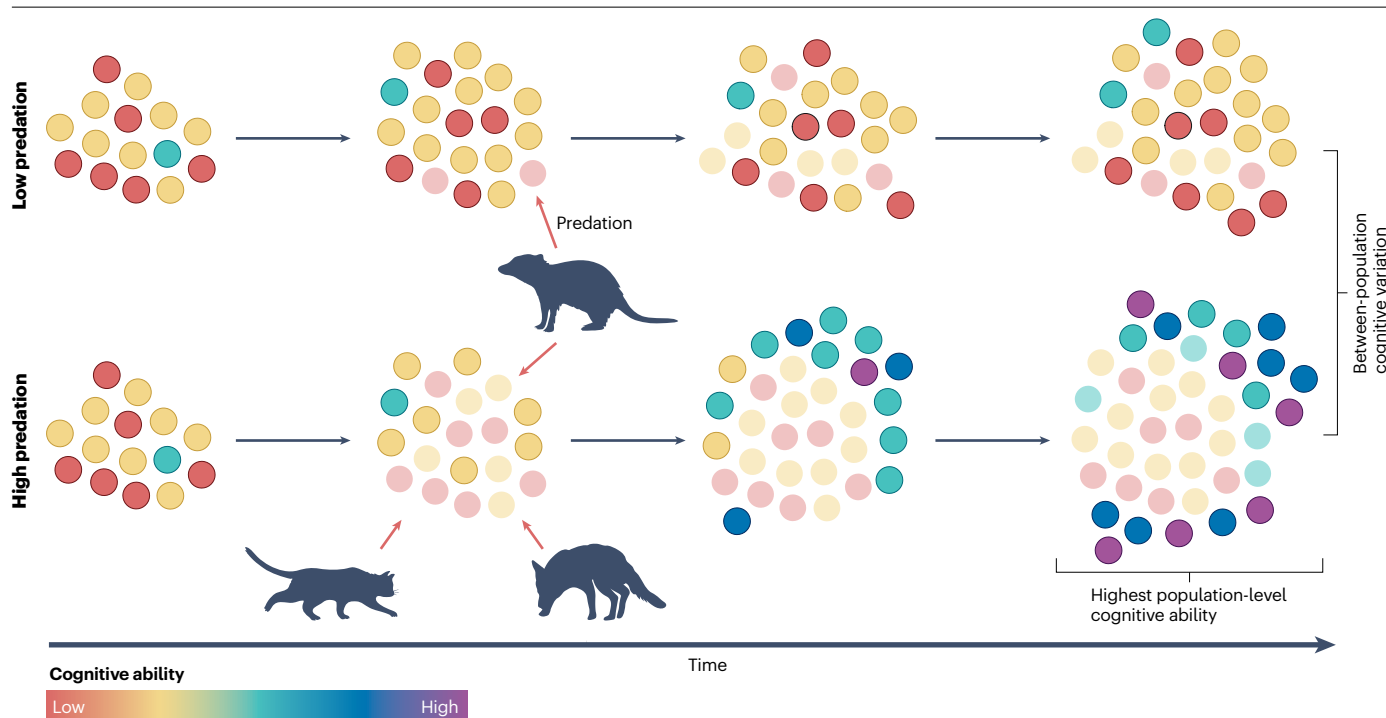
Predation might help to maintain cognitive variation at various biological levels, as species, populations and individuals co-exist with heterogeneous predator communities. Prey populations living alongside diverse predatory guilds should be under stronger selection for enhanced cognition as they must recognize and respond to many different predatory cues predicting predation risk<sup>42</sup>, whereas populations facing a single predator will have fewer cues to remember. By this logic, the cognitive demand of predator avoidance is substantially higher for prey in ecosystems with diverse predatory pressures. Thus, geographic and temporal variation in predatory regimes should correspond with cognitive variation in prey (Fig. 2).

## Evolutionary trade-offs and enhanced cognition

Enhanced cognitive capacity confers adaptive advantages, but maintaining a large brain is metabolically costly<sup>43,44</sup>. Brain size is influenced by life-history traits owing to energy allocation trade-offs. For example, investment in reproduction in terms of offspring number, offspring size and frequency of reproduction can limit the energy available to

maintain brain tissue, particularly in females<sup>45</sup>. Likewise, the metabolic costs of other life-history traits such as longevity, growth rate and reproductive age could similarly be traded off against brain size, as part of an overall life-history strategy<sup>46–50</sup>. Under some scenarios, investment in brain tissue might not reduce mortality risk as much as investment in muscle tissues directly linked to survival behaviours, such as escape speed<sup>51,52</sup>. The development of physical defences also reduces the need for large brains<sup>26</sup>. A similar scenario exists for species that rely heavily on crypsis to avoid detection by predators, or species that use pursuit deterrent signals that honestly signal either physiological performance or simply that a predator has been seen by potential prey<sup>53</sup>. In some species, anti-predator strategies might not require a high cognitive load (for example, if fitness is largely independent of cognition).

Conversely, cognitive evolution can be driven without increased predation pressure; for example, guppy progeny translocated to low-predation streams display increases in brain size<sup>54</sup>. Generally, the strength of predation on cognitive abilities seems to be species-dependent, which is probably a product of the fitness pressures of a given species' life-history strategy<sup>51,55,56</sup>. Some species subject to very intense predation risk and mortality might instead be selected for fast life-history strategies, allowing reproduction before mortality and limiting the development of large brains. Understanding which species and traits correlate with enhanced cognitive abilities is a promising avenue for understanding evolutionary trade-offs driven by predator–prey interactions.



**Fig. 2 | Predation as an evolutionary force shaping cognitive variation.**

Enhanced cognition is associated with increased survival and greater predation risk<sup>22,38</sup>, suggesting that predation may select for cognitive performance. The two populations depicted experience low and high predation pressure. Individuals are represented by the circles in each population, and individuals who have been predated are faded out. Selection for cognitive abilities is higher where predation pressure is diverse and amplified, promoting enhanced cognition and driving

divergence from the low-predation populations. Cognitive variation emerges as predators select for prey cognition differently within and between populations. Predatory functional complexity is therefore expected to generate and maintain cognitive variation among and between prey populations. The same pattern is predicted for predators, but instead of predatory functional complexity, the functional complexity of prey species is likely to drive cognitive variation.

## Mechanisms of selection on prey cognition

Given that prey are at almost constant risk of being predated, they have evolved a suite of anti-predator traits and behaviours<sup>53</sup>. Anti-predator behaviours are behavioural adaptations that allow prey to avoid detection, evade attack or fight back<sup>57</sup>. The ability of prey to remember and interpret predator cues (olfactory, chemosensory, visual, vibrational and auditory), identify and recall areas of high predation risk, learn from previous predator encounters and learn from conspecifics about predators are all cognitively demanding challenges<sup>19,33</sup>. This iterative learning process is constantly repeated through an individual's life, a process known as Bayesian updating. Individuals who are more readily able to integrate and react to new information are likely to have an advantage when it comes to avoiding predation compared with those unable to integrate information as effectively<sup>34,58–60</sup>.

Prey are also required to use past information to predict future events and use multiple sensory modalities to do so – correctly predicting the location and behaviours of a predator is likely to reduce mortality<sup>61</sup>. Similarly, individuals who can remember and learn when to be vigilant, when and where to flee or hide, and know when to not invest in these costly behaviours are likely to be selected for under predation risk. Cognitive traits such as pattern recognition, learning ability, causal understanding and spatial memory are all likely to be highly advantageous under these conditions<sup>34</sup> (see Supplementary Table 1 for predictions under the PIH). If enhanced cognitive abilities significantly reduce predation risk, as some have suggested<sup>15,38,59</sup>,

enhanced cognition should provide a clear selective benefit when predation pressure is high.

## Mechanisms of selection on predator cognition

Predators have evolved behavioural strategies to maximize their hunting success, including matching the diel activity patterns of their prey, recognizing prey cues and using hunting strategies that exploit the morphological vulnerabilities of their prey<sup>62–65</sup>. Predators therefore also depend on cognitive traits such as pattern recognition, learning ability and spatial memory, among others. The ability to recognize prey search images from the visual environment should allow predators to better detect and hunt prey<sup>66</sup>. Predators who are better able to recognize a range of prey cues (such as scents) and remember prey congregation locations or areas of previous hunting success are more likely to be successful hunters.

Selection for cognitive capabilities might be strongest in predators hunting diverse guilds of different prey types, where predators must remember and integrate a wide variety of cues providing information about prey<sup>67,68</sup>. This could also apply to mimicry complexes, which can be highly diverse with varying levels of protection. For example, one mimicry complex in Australia contains 140 potential mimics from 4 arthropod orders that include ants, spiders, wasps, bugs and tree hoppers. These mimics range from non-defended Batesian mimics to heavily defended Mullerian mimics with protection that includes, to a varying degree, spines, cuticle thickness, stinging, mandible size,

poison gland size and aggressive behaviour or communal attack<sup>69</sup>. Mimicry complexes can thus represent polymorphic states, adding to the cognitive load required to navigate a landscape of highly variable prey, although researchers' understanding of these systems is still developing. Nevertheless, in the context of predator cognition, many predators sample potential prey (mimics and models) and need to learn which traits signal distastefulness and at what point a signal might not be accurate or reliable, and whether or not to avoid these potential prey in the future or, in some cases, how best to get past their defences<sup>70–72</sup>. Other cognitive traits such as the ability to anticipate and respond to anti-predator behaviours might help predators to successfully subdue prey<sup>73</sup>. Enhanced cognitive abilities are therefore probably highly beneficial to predators in many contexts<sup>20,74</sup> (Supplementary Table 1).

Selection for cognitive abilities could also be stronger when prey is scarce, as each failed hunt has a higher impact on an individual's probability of survival, as outlined in the marginal value theorem. When prey resources dwindle, predators should make optimal decisions around what and where to hunt, maximizing their fitness in the process. Inevitably, no foraging is optimal and predators are likely to be selected for their abilities to recognize and respond to prey resource limitation and potentially prey switch to a more abundant option<sup>75</sup>. Then, individuals who best integrate information about food scarcity and adjust their hunting behaviour accordingly are likely to survive.

Some prey species are challenging to predate. Hunting dangerous prey, for example, could be more cognitively challenging than hunting less risky prey. Dangerous prey can injure predators, potentially reducing their capacity to hunt in the future or even causing death<sup>76</sup>. This added fitness cost probably selects for predators with knowledge of how to hunt dangerous prey, the ability to plan ahead and the judgement to abort risky hunts. Dangerous prey can also drive teaching and social learning of predatory strategies. For example, meerkats (*Suricata suricatta*) teach their young how to remove the stingers of their venomous scorpion prey<sup>77</sup>. Moreover, generalist predators – which hunt across different functional groups – probably face greater cognitive demands than specialist predators<sup>78</sup>. Generalists are challenged to remember the cues and locations of different prey, store multiple search images and have memorized distinct hunting strategies, such as solo hunting for small prey and cooperative hunting for large, dangerous prey. Additionally, socio-cognitive traits in cooperative, group-hunting predators could also be subject to selection, because individuals that successfully hunt prey cooperatively should derive fitness benefits<sup>79</sup>. To hunt as a group, predators must act on uncertain information, making both individual and collective probabilistic estimations about the likely decisions and behaviours of both their cooperators and prey species<sup>80</sup>. Cooperative hunting is at the interface of the PIH and the SIH because there should be selection both for the ability to track particular individual conspecifics and their roles in a hunt<sup>81</sup>, and for the ability to learn from past experience based on how prey behave in particular circumstances and thereafter make adjustments in future hunts. On the whole, the cognitive abilities of carnivores (and, probably, predators in general) are poorly understood<sup>5</sup>.

## How predation shapes cognition

Cognition varies greatly between individuals, populations and species. Testing how predation shapes cognition across these hierarchies requires asking distinct questions with different systems and methods. In this section, we outline questions central to our understanding of predation's role in the evolution of cognitive variation (Supplementary Table 1).

## Methods for testing the PIH

Historically, predation pressure has been considered almost impossible to estimate in most environments<sup>15</sup>. However, over the past two decades, advancements in field-based methods – such as camera traps<sup>82</sup>, animal-borne cameras, audio recorders and biologgers<sup>83</sup> – have substantially improved scientists' ability to characterize predator behaviour and estimate the complexity of predation prey experience. These methods allow for the tracking and detailed behavioural analysis of individuals during predator–prey interactions, which can be correlated with cognitive performance<sup>22</sup>. Furthermore, researchers can now explore predatory complexity in deep time, facilitated by the development of ancient predator–prey network simulations<sup>84–87</sup>. These advances have enabled quantification of the number of predators with which prey species co-existed and allowed researchers to move towards more direct proxies for predation pressure and complexity.

Functional macroecology has also rapidly advanced in the past decade, enabling modelling of historic and current-day predatory complexity based on the traits of predators and prey<sup>88</sup>. Although these methods provide simple measures of predation (such as the number of predators present in a biome), they lack both spatial and temporal nuance of realized interactions. The modelling of ancient predator–prey networks allows the quantification of predator–prey coexistence history and the calculation of the functional complexity of the predator guild. Correlating neuroanatomical proxies of cognitive ability with social or environmental variables<sup>4,14,89</sup> has been central to understanding cognitive evolution. With these advances in network and functional ecology, researchers can now use metrics of past ecological communities (such as the number of predators and the functional richness of the predator guild) alongside neuroanatomical data to explore how predation has driven cognitive evolution through time. In summary, testing how predation shapes cognitive variation is more possible now than ever before.

## Cognitive variation among species

The modelling of now extinct predator–prey interactions<sup>87,90</sup> through network analysis and interaction inference methods can help researchers to understand how interspecific cognitive variation evolved<sup>85</sup>. Ecological networks are further complemented using the traits of these predators to ask whether prey who experienced functionally complex predation (for example, both pounce-pursuit and ambush predation styles) evolved larger brains. This exploration can be done at the macroevolutionary scale, using phylogenetic comparative approaches<sup>91,92</sup>, which are powerful tools for testing for the correlational evolution of traits. We hypothesize that experiencing predation from multiple predators with distinctive hunting styles presents a greater cognitive burden for remembering and responding to cues and patterns than a single predator, or even multiple predators with functionally similar hunting styles. This prediction is supported by the multi-predator and predator archetype hypotheses<sup>68,93</sup>, which state that prey individuals with multiple functionally similar predators generalize anti-predator responses across those predators. If prey live with a functionally diverse set of predators, anti-predator behaviour generalization could be more cognitively challenging. Examples of this strategy can be found in prey species that develop unique anti-predator behaviours for functionally diverse predators, such as sooty mangabeys (*Cercocebus atys*) having distinct strategies for reptilian and mammalian predators<sup>94,95</sup>. Developing strategies for the avoidance of multiple predators is likely to be more cognitively taxing

## Glossary

### Bayesian updating

The process by which individuals integrate new information to update existing knowledge.

### Causal understanding

The ability to understand and integrate cause and effect.

### Cognitive performance

The success with which an animal solves a problem or learning task, including memory. Measures of cognitive performance might include latency to problem-solving or the number of mistakes leading to problem-solving.

### Giving-up density

A foraging experimental procedure that measures foraging decisions in individuals under varying risk conditions.

### Inhibitory control

An individual's ability to resist automatic urges.

### Innovation

The development of a novel or modified behaviour.

### Learning

A change in behaviour dependent on prior experience.

### Marginal value theorem

A model that describes how individuals make optimal foraging decisions under depleting resources.

### Predation pressure

The intensity of influence that predators exert on prey, typically expressed as the risk of being killed or attacked by a predator. Predation pressure encompasses both the consumptive (being killed) and non-consumptive (fear and avoidance) effects of predation.

### Predator–prey interactions

A trophic interaction in which predators hunt prey and prey use strategies to reduce their likelihood of being killed by a predator.

### Spatial memory

An individual's ability to remember locations and navigation pathways.

### Telencephalon

The most developed and anterior part of the forebrain, which has an important role in cognitive functions.

than for a single predator or predator archetype. This predatory functional diversity should thus select for prey with larger brains as prey should require anti-predator behaviours for each predator species<sup>96,97</sup> (Supplementary Table 1).

Similar patterns could be seen in predators that historically hunted many prey species, each of which had unique anti-predator strategies. Broadly foraging animals are hypothesized to exhibit enhanced cognitive abilities owing to the cognitive pressures associated with hunting many functionally diverse prey across large and diverse habitats<sup>6</sup>. Using the network methods detailed here, one could extract the number of prey species a predator species evolved alongside and quantify their functional traits. This quantification thereby facilitates the testing of whether functional complexity in historically hunted prey predicts the evolution of larger brains. A logical hypothesis follows – that diverse prey guilds select for larger brains or enhanced cognitive abilities; however, the presence of one particularly challenging prey species (dangerous or large-bodied species) could predominantly drive this pattern given the enhanced selective pressure placed on predators by potentially lethal prey. Evaluating links between prey functional traits and measures of predator cognition is a promising avenue for improving understanding of how predator–prey interactions shape cognitive variation.

## Predation, cognition and development

Researchers have a limited understanding of how predation shapes the development of cognitive phenotypes. In accordance with how the social environment can shape the cognitive development of individuals as they age (individuals living in larger groups exhibit enhanced cognitive abilities and improved fitness<sup>13</sup>), predation might have a comparable effect during developmentally sensitive time periods.

Initial effects could begin in the womb, in response to the mother's environmental risk of predation. Maternal effects could influence the offspring's cognitive and behavioural phenotype<sup>98,99</sup>. Subsequent effects could be triggered during early social development in response to the local environment, in this case potential exposure to predators and increased predation risk. Juveniles, being the most vulnerable to predation, might develop cognitive abilities that promote predator recognition and avoidance. We suggest that predation risk in early life might drive the development of enhanced cognitive abilities as a mechanism to improve predator detection and avoidance, similar to how it influences physiological performance required for rapid escape<sup>52</sup>.

Additionally, some precocial species appear to be born with appropriate anti-predator responses towards mimicked cues of a predator<sup>100</sup>; however, it is unclear how this is shaped by transgenerational plasticity or how plastic these responses might be to shifts in predatory regimes. At the species level, one might expect some interplay between play behaviours during development, which correlate with large brains and the development of skills that are used to avoid predation<sup>101</sup>. Quantifying individual cognitive abilities from juvenile through to adulthood under varying predation regimes will provide insights into which cognitive traits are favoured in differing predation regimes, elucidating the developmental effects of predation on cognitive abilities. This approach can be applied to both predators and prey; causal relationships could be identified through the cross-fostering of eggs in avian systems, or a split-clutch allocation of eggs in reptilian systems across areas of high and low predation regimes. Furthermore, some systems, such as lizards, are excellent models for manipulating maternal effects and the early social environment to study the effects of predation risk on cognitive ability and behaviour<sup>102</sup>.

## Cognitive ability variation across hierarchies

Over the past 20 years, cognitive ecologists have become increasingly interested in individual cognitive variation. By unpacking the drivers of individual variation in cognitive abilities, researchers can address questions that tackle both proximate and ultimate questions surrounding cognitive evolution. This could be done by combining cognitive test batteries with predation risk experiments such as giving-up density with predator cues<sup>103,104</sup> or playback experiments for prey<sup>105</sup>. Combining predation risk experiments and cognitive testing allows for investigation of the mechanisms underpinning the correlations between enhanced cognitive abilities and improved survival in the wild. Both islands and conservation havens that vary in the number of predators are fruitful systems for these questions, where one could ask how predation risk shapes the spectrum of intraspecific diversity in cognitive abilities. Observational methods such as GPS tracking and animal-borne video can help to identify predation success. Initial study species should generally be solitary where possible to disentangle individual from group hunting success.

Variation in cognitive abilities among prey populations might emerge through context-specific predation pressures. Islands are ideal systems for studying how predation influences cognitive evolution

because they offer simplified trophic networks, and they can vary naturally in predator and prey species distribution<sup>29–31</sup>. Additionally, there is a growing number of fenced conservation reserves, where prey are either separated from their predators or live under different predator treatments (such as native predators only or predator-free settings)<sup>106,107</sup>. These fenced reserves can function as islands for comparison. Ideally, intraspecific cognitive comparisons should occur across areas of similar environmental complexity to control for environmental heterogeneity. Lower dietary diversity on smaller islands have been hypothesized to contribute towards reduced brain size and cognitive performance in predators<sup>108</sup>. Fewer species to hunt could equate to fewer cognitive challenges on islands compared with the mainland, thereby reducing the strength of cognitive selection. This reduced strength of selection could result in mainland populations exhibiting enhanced cognitive abilities compared with those on islands.

## Outlook

A key issue in psychology, behavioural ecology and evolutionary biology is determining the factors governing cognitive variation across the animal kingdom, including at the species, population and individual levels. Well-established hypotheses such as the SIH and the EIH provide important explanatory power to account for variation in cognitive ability, but they fall short in some systems and circumstances. We describe the PIH and formalize the role of predator–prey interactions as an additional driver of cognitive variation.

There are many promising avenues for studying predation's role in shaping cognitive ability. Researchers looking to test the PIH at the population and individual levels should prioritize systems that allow for the direct manipulation of predation pressure, such as predator enclosures, islands, developmental approaches or lines of animal models that vary in generational exposure to predation. These approaches offer the best chance at causal inference and should exclude altering variables predicted to shape cognition in other hypotheses (such as group size or environmental complexity). Further, linking cognitive performance with anti-predator behaviour or predatory success is an important step for understanding the mechanisms driving predation's influence on cognitive abilities. For researchers interested in studying the PIH at the species level, we recommend leveraging approaches from macroevolution and network and functional ecology to recreate prehistoric or ongoing predation using metrics such as the number of predators or prey per species in a given system. These metrics can be correlated with cognitive ability, brain size or anatomy to provide insights into predation and cognitive evolution. Pertinent first comparisons might involve comparing island and mainland species or areas with low (such as Australia) and high (such as Africa) predator species richness.

As highlighted, the PIH overlaps with the SIH and the EIH. Therefore, understanding when predatory, social and ecological drivers might synergistically drive cognitive evolution is an important step forward. For example, both the PIH and the SIH predict group-hunting predators to exhibit enhanced cognitive abilities compared with non-group-hunting predators, as first testing has documented<sup>109</sup>. Further, the cultural intelligence hypothesis, which specifically posits that frequent opportunities for social learning should improve learning ability<sup>110</sup>, also exhibits strong overlap with the PIH. Combined, the cultural intelligence hypothesis and the PIH would predict that predators that learn to hunt socially should exhibit enhanced cognitive abilities. Under the PIH and the EIH, predators who hunt across variable habitats might be expected to exhibit enhanced cognitive abilities compared

with those that hunt in single habitats. Finally, there might exist an intersection between all three hypotheses, where cognitive abilities might be strongest in predators who cooperatively hunt a diverse variety of prey in highly variable habitats. For researchers interested in taking a unified approach to studying cognition we recommend intraspecific comparisons, likely model animal lines that have been exposed to varying degrees of predatory, social and environmental complexity. Given the challenges of testing all hypotheses at once, other systems could be challenging and create noise, and the existence of an ideal island system where the same species live at varying densities, among various predators and across a gradient of environmental complexity is unlikely. Unifying these hypotheses is an essential research direction for understanding the drivers of cognitive evolution.

We have detailed a series of testable predictions and discussed how technological advances in monitoring behaviour in the wild might be leveraged to improve our understanding of the factors driving differences in cognitive ability in the context of predator–prey interactions. In formalizing the PIH, we hope this will stimulate new research to address variation in cognitive ability that is currently unaccounted for.

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## Author contributions

E.I.F.W. and B.J.A. developed the concept. E.I.F.W. wrote the original draft. All authors contributed substantially to the revisions and final text.

## Competing interests

The authors declare no competing interests.

## Additional information

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